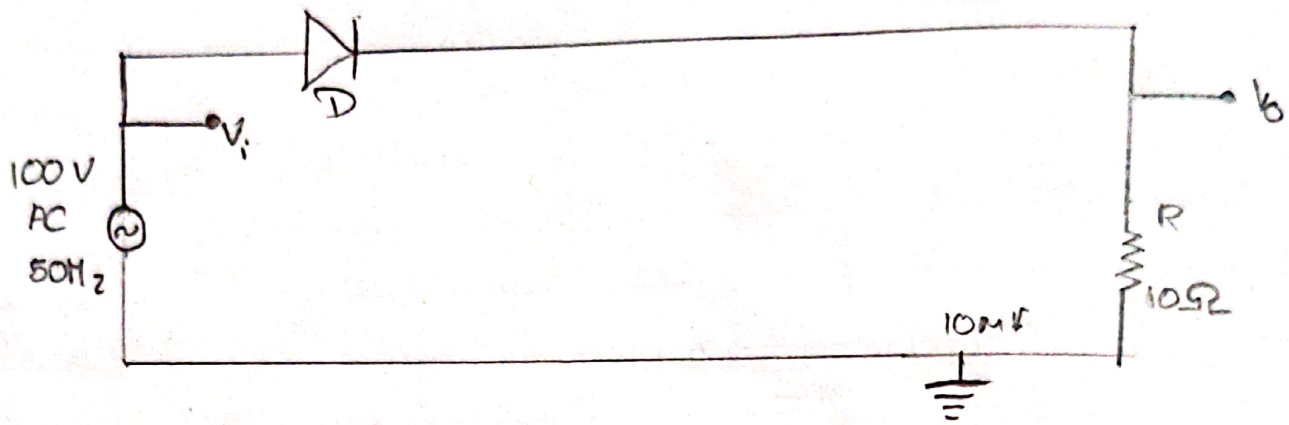
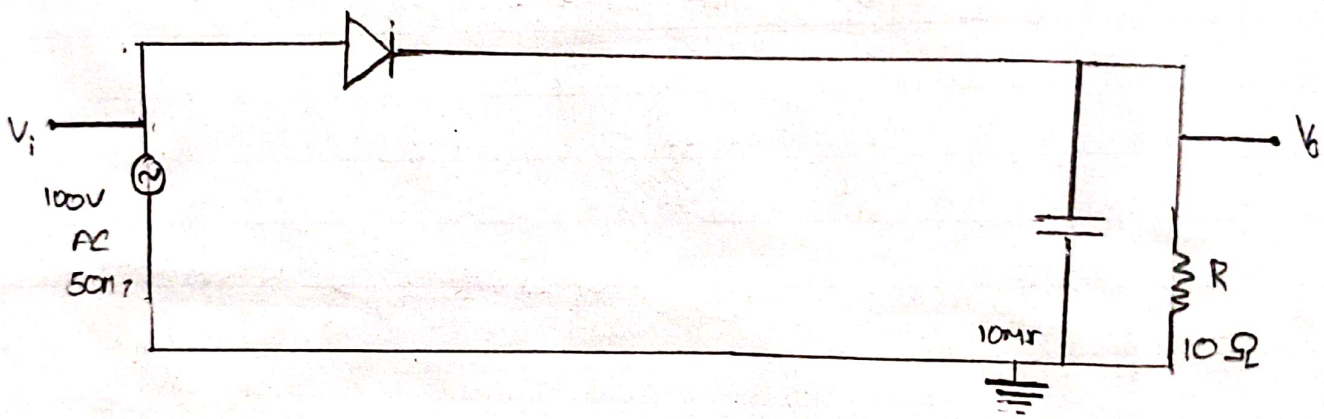
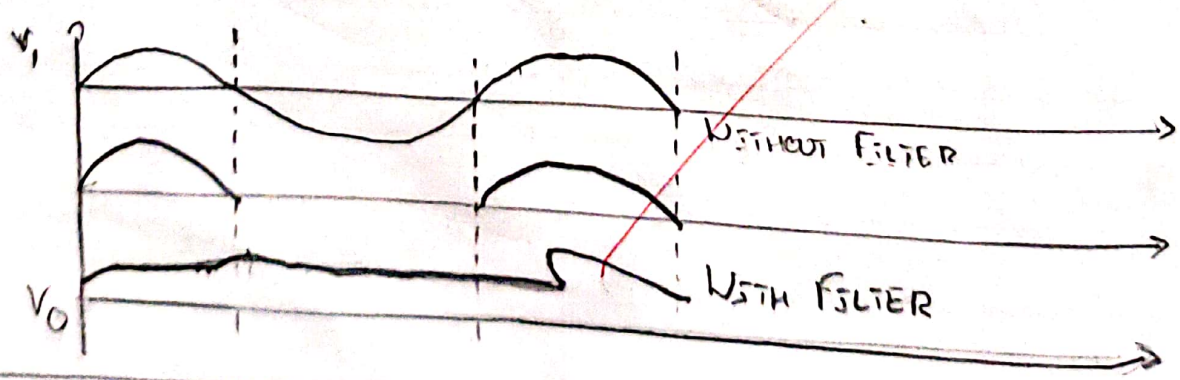
WITHOUT FILTER



WITH FILTER



Input Voltage		Output Voltage			
		without filter		with filter	
$V_m(t)$	t (ms)	$V_u(V)$	t (ms)	$V_m(V)$	t (ms)
100V	20ms	100V	100 ms	100V, 85V	5ms, 25ms



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HALF WAVE AND FULL WAVE RECTIFIER.

AIM:-

To simulate the following circuits proteus:-

i) Half wave rectifier.

ii) Full wave rectifier.

APPARATUS REQUIRED:-

Laptop with Proteus software.

THEORY:-

A rectifier is a circuit that converts alternating current (AC) into direct current (DC).

HALF WAVE RECTIFIER:-

Half wave rectifier conducts only during positive half cycle. During positive half cycle the diode conducts and the output voltage is equal to the input voltage. During negative half cycle, the diode does not conduct and the output voltage is equal to zero.

FULL WAVE RECTIFIER:-

Full wave rectifier conducts during both positive and negative half cycles. During positive half cycle, the output voltage is equal to input voltage during negative half cycle. The output voltage is equal to negative of the input voltage. The output is of the

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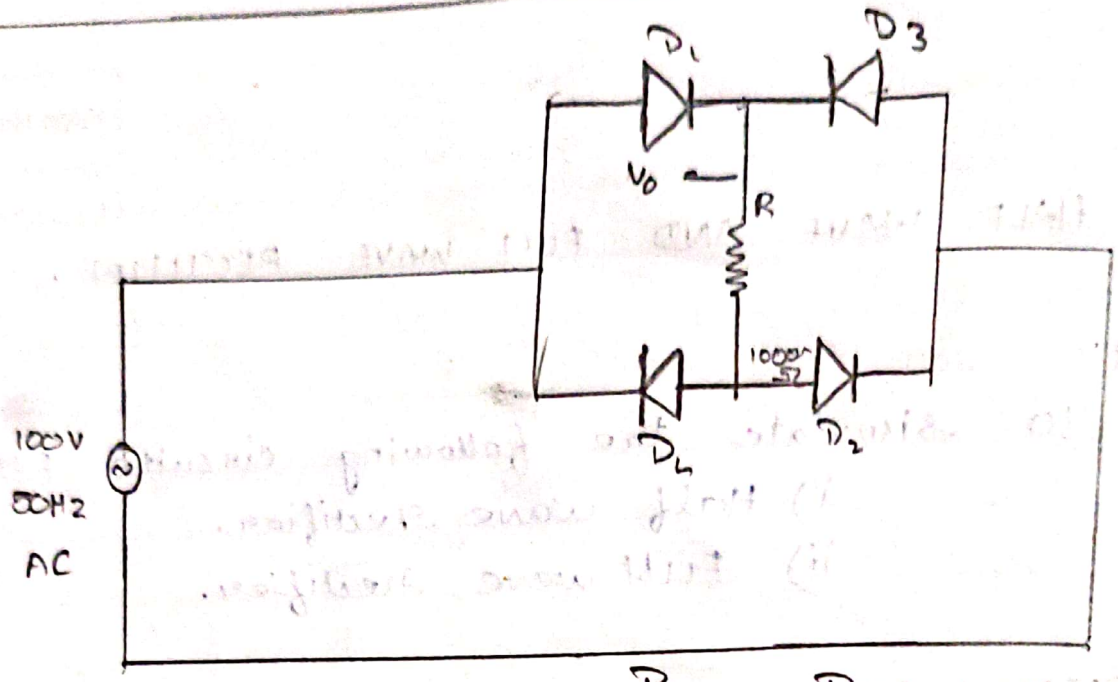
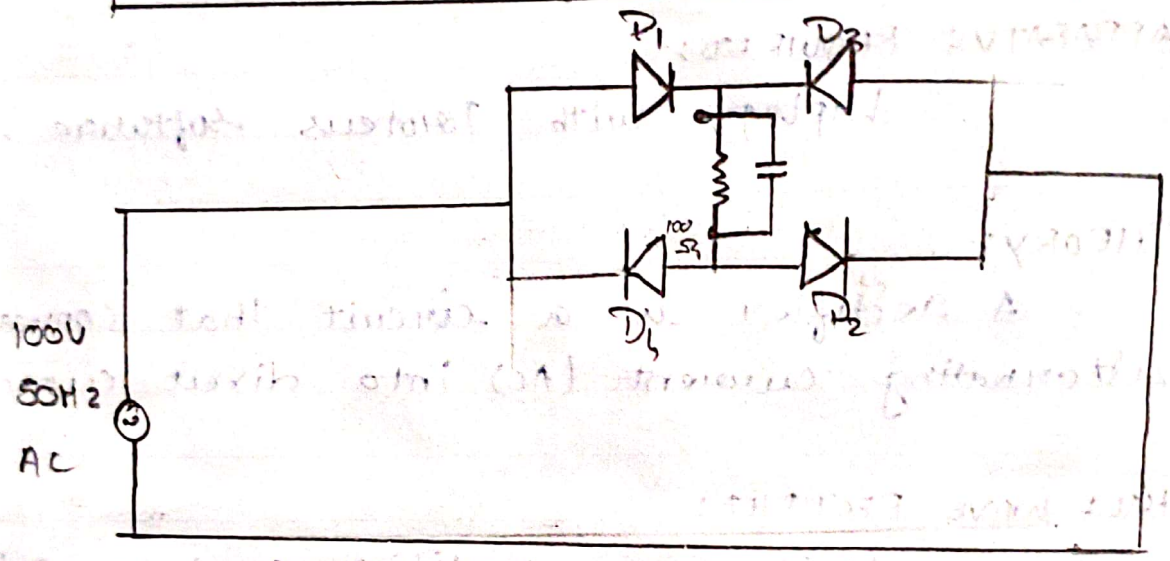
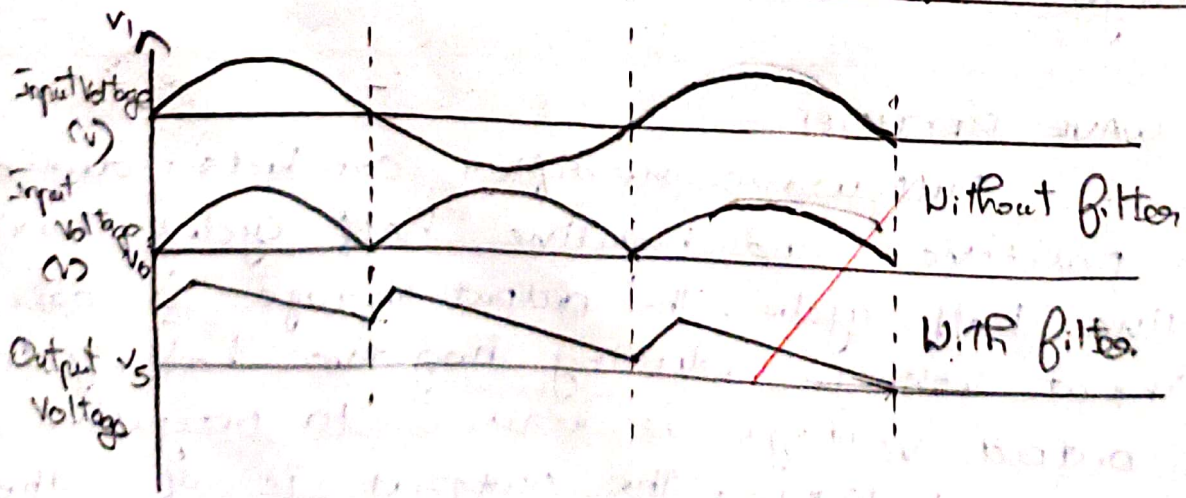




Input Voltage		Output Voltage			
		without filter		with filter	
Vm(V)	t (ms)	Vm(V)	t(ms)	Vm(V)	t(ms)
50V	20ms	50V	5μs	100V, 46V	5μs, 1ms



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rectifier circuit is impure DC.

In order to get, pure DC the output is filtered by a capacitor filter.

PROCEDURE:-

1. Drag the required components from the Proteus library.

2. connect the components as per the circuit diagram.

3. Measure the input and output voltages and note in tabulations.

4. Draw the graph for input and output voltages.

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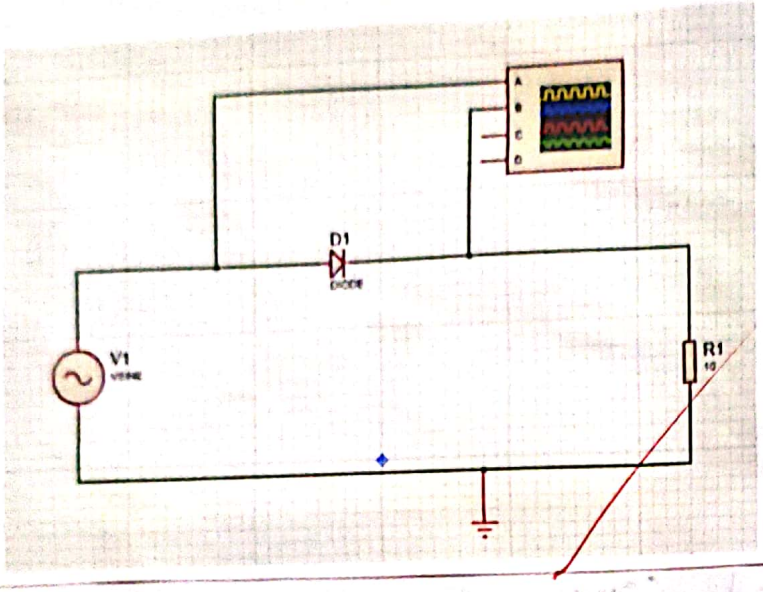

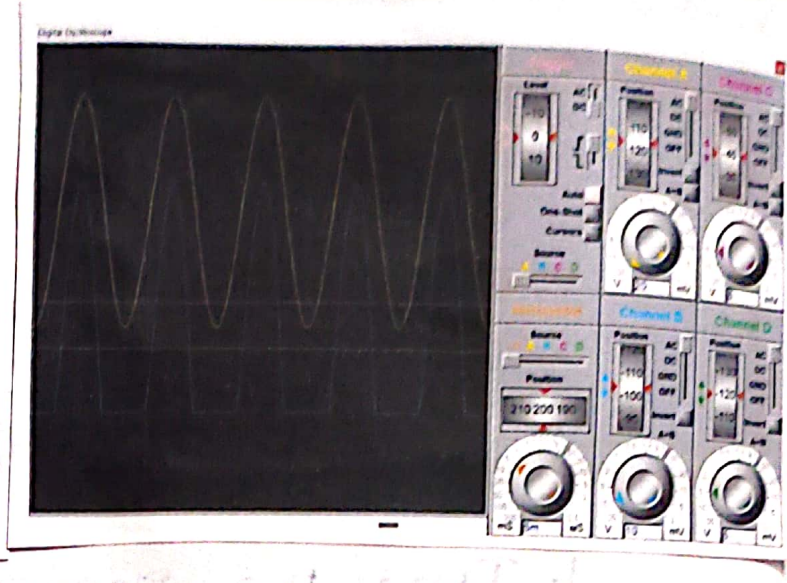

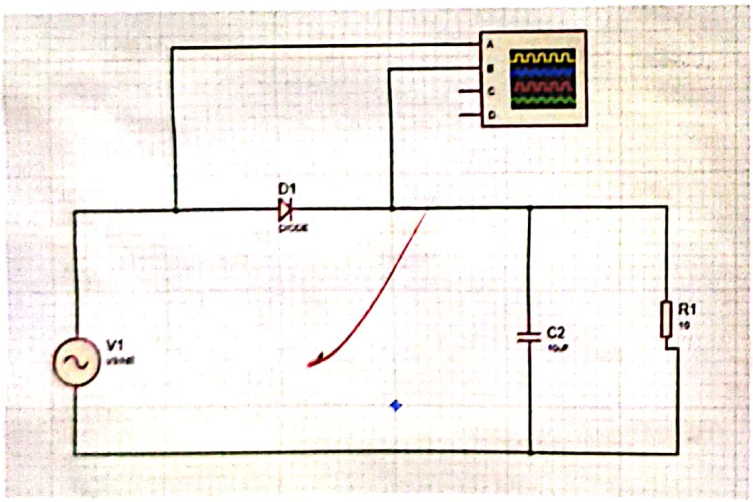

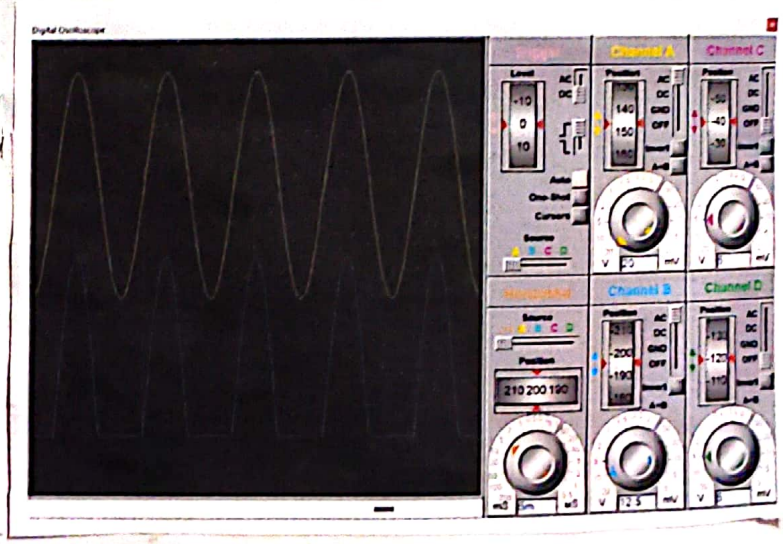

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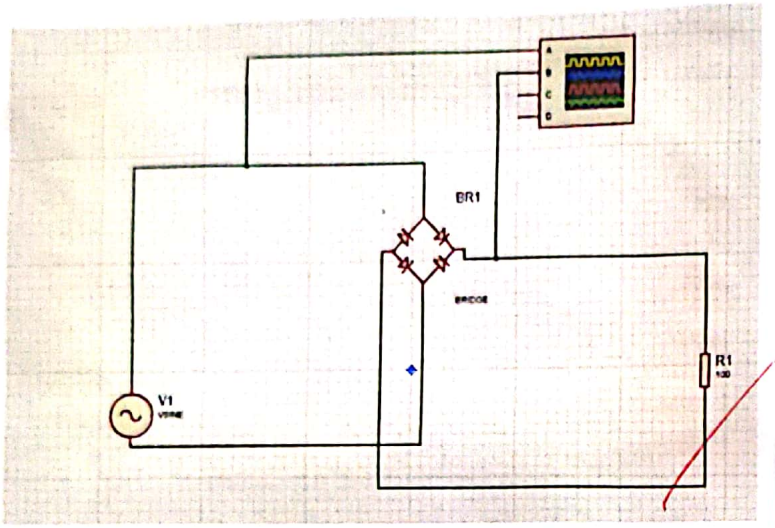

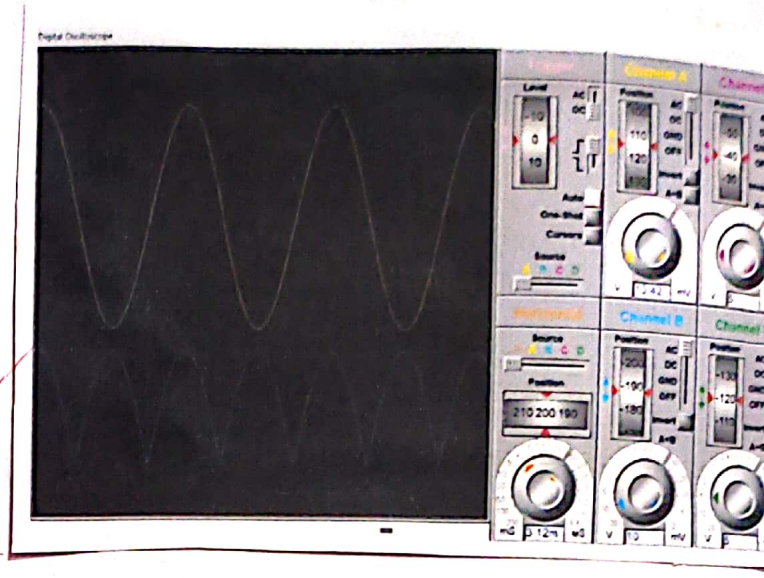

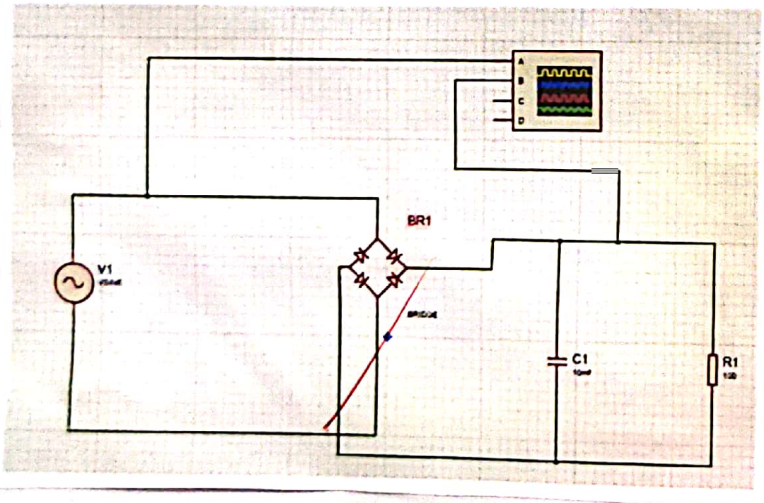

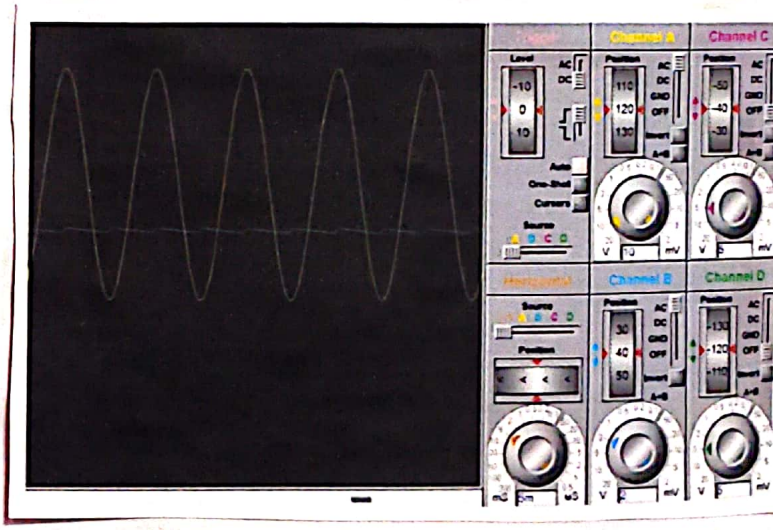

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